

Practice Questions: Spanning, Linear Independence, and Subspaces

Part A: Conceptual Questions

1. In your own words, explain the difference between:
 - a) A *spanning set* of vectors.
 - b) A *linearly independent* set of vectors.
 - c) A *basis* of a subspace.
2. Why is it useful in AI/ML to have a basis for a subspace instead of a redundant spanning set? Give one example.

Part B: Computation Problems

1. Span & Linear Dependence Check

Consider vectors in $\mathbb{R}^3$:

$$v_1 = (1,2,1), v_2 = (2,4,2), v_3 = (0,1,1).$$

- a) Do v_1, v_2, v_3 span $\mathbb{R}^3$?
- b) Are they linearly independent? Justify.
- c) Find a basis for the subspace spanned by $\{v_1, v_2, v_3\}$.

2. Subspace in AI Context

Suppose each vector represents a **document embedding** in $\mathbb{R}^3$:

$$d_1 = (1,0,1), d_2 = (0,1,1).$$

- a) Find the subspace spanned by $\{d_1, d_2\}$.
- b) Is the vector $q = (1,1,2)$ (a query) in this subspace? Show work.
- c) Interpret what it means (in words) if q belongs to the span of $\{d_1, d_2\}$ in terms of *document similarity*.

3. Application: Image Compression Idea

Suppose images are represented as vectors in $\mathbb{R}^4$. You are given two orthogonal vectors:

$$u_1 = (1,0,0,1), u_2 = (0,1,1,0).$$

- a) Show that $\{u_1, u_2\}$ is linearly independent.
- b) What is the dimension of the subspace they span?
- c) Explain how restricting to this subspace could represent a **low-rank approximation** in AI (e.g., compressing image data).

Part C: Challenge (Optional)

Let $S = \{(1,1,0,0), (0,1,1,0), (0,0,1,1)\}$ in $\mathbb{R}^4$.

- Find the dimension of $\text{span}(S)$.
- Construct a basis for $\text{span}(S)$.